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M4D: Manifold entanglement for robust decision tree NIDS in medical IoT

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ABSTRACT

Decision trees are increasingly favored in lightweight Network Intrusion Detection Systems (NIDS) for Medical IoT (MIoT) due to their efficiency and interpretability, which are critical for resource-constrained healthcare environments. However, these models remain vulnerable to adversarial attacks that can manipulate network traffic features and distort decision boundaries, resulting in dangerous misclassifications of malicious activity. Existing defense mechanisms often lack robustness, particularly when confronted with unseen attack strategies. To address these challenges, we introduce the Manifold-based Defense framework (M4D), tailored for decision tree-based NIDS in MIoT. M4D entangles projected latent spaces to enforce interdependencies among traffic features, thereby impeding low-cost adversarial manipulation. Additionally, it incorporates cross-manifold mixup augmentation to further smooth decision boundaries and enhance robustness against adversarial perturbations. Extensive evaluation on two recent MIoT network intrusion datasets, CICIoMT2024 and ICU, under both benign and five diverse adversarial attack scenarios, demonstrates that M4D consistently maintains over 90% accuracy, surpassing six benchmark defense methods. These findings underscore the potential of M4D for enhancing the security and reliability of anomaly detection in critical medical IoT networks.

1. Introduction

The Internet of Things (IoT) has expanded from smart homes and industrial automation to safety-critical domains such as connected healthcare. This growth has introduced a fundamental security tension: IoT deployments often operate at massive scale, with heterogeneous devices and long lifecycles, yet many endpoints are difficult to secure through traditional means. In practice, devices may be resource constrained, vendor-locked, intermittently connected, or maintained under strict regulatory and operational constraints. Consequently, timely patching and host-level instrumentation are frequently limited, incomplete, or infeasible—a gap that adversaries can exploit for stealthy, persistent intrusion. These constraints make network-level defense essential. Network Intrusion Detection Systems (NIDS) provide centralized monitoring and protection across diverse device ecosystems, without requiring changes on each endpoint. This is particularly important in Medical IoT (MIoT), where patching legacy devices may be rare or impossible due to certification requirements and clinical operational constraints. In such settings, NIDS is not merely an additional control; it often becomes the primary (and sometimes the only) realistic security layer that can be deployed quickly and scaled broadly.

Among Machine Learning (ML) approaches for NIDS, Decision Trees (DTs) remain especially attractive. DTs combine simplicity, interpretability, and computational efficiency, making them practical for deployment on IoT edge gateways and monitoring infrastructure (Podgorelec et al., 2002). Their split-based hierarchy enables real-time inference with minimal overhead, while producing human-readable rules that align with Security Operations Center (SOC) workflows. Unlike many black-box models, DT decisions can be inspected to identify which feature regions triggered an alert and to trace these conditions back to meaningful network and logging semantics (e.g., protocol behavior, flow duration, byte/packet patterns). This transparency supports trust and adoption: analysts can validate alarms, explain decisions to stakeholders, and refine operational playbooks. DT-based NIDS also reports strong accuracy on public cybersecurity datasets, often exceeding 99% in standard (clean) evaluation settings (Dadkhah et al., 2024; Altulaihan et al., 2024).

Yet, high clean accuracy does not necessarily imply robustness: these evaluations typically measure performance on benign traffic drawn from the same distribution as the training data, whereas real attackers deliberately manipulate inputs to violate this assumption. Consequently, the very reliance on learned decision boundaries exposes

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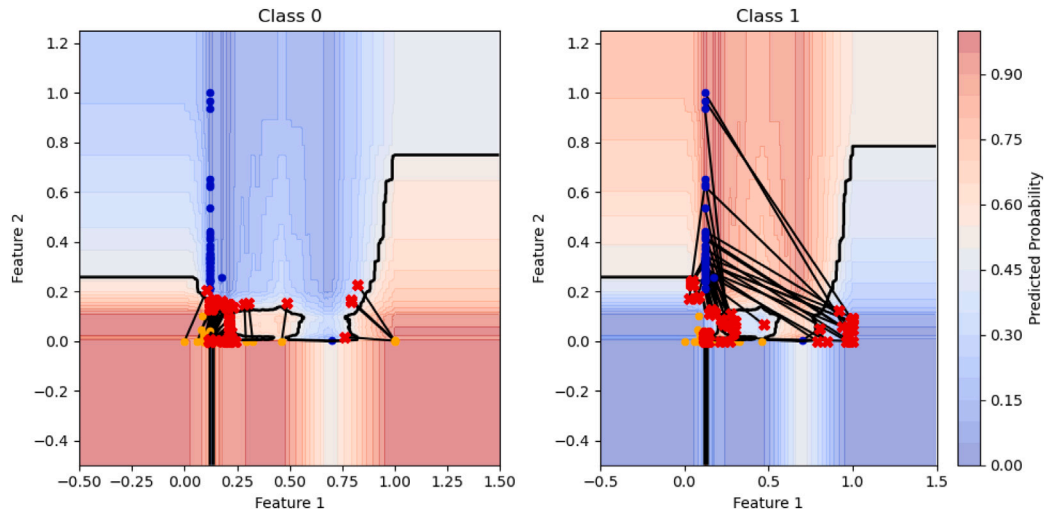


Fig. 1. 2-D illustration of adversarial attacks: colored background indicates the model's predicted probability for class i ; blue circles are class 0 samples; orange circles are class 1 samples; and red crosses mark adversarial examples targeting class i .

DT-based NIDS to adversarial attacks. Adversarial examples are subtle, strategically crafted perturbations that steer inputs toward regions where the classifier becomes unreliable, causing misclassification (see Fig. 1) (Vos and Verwer, 2023). This vulnerability is tightly connected to domain shift: adversarial perturbations can push samples away from the training distribution, where the learned split rules no longer reflect stable behavior (Vos and Verwer, 2023). The challenge is compounded by attack diversity. Adversaries can exploit different constraints, objectives, and knowledge levels, yielding a virtually unbounded space of strategies that makes it impractical to pre-train defenses for every scenario (Cohen et al., 2024).

Although adversarial attacks are broadly applicable across IoT networks, their impact in MIoT is both more severe and less explored in the literature. MIoT deployments increasingly rely on ML models not only for network protection, but also across clinical workflows such as diagnosis support, treatment recommendation, and continuous patient monitoring. In this context, a successful adversarial attack can have direct safety consequences: false negatives may suppress critical alerts and miss life-threatening conditions, while false positives may trigger unnecessary interventions by erroneously indicating conditions that are not present (Gisselsson, 2022). These misclassifications can cascade into delayed treatments, unnecessary procedures, and heightened patient anxiety, ultimately posing direct threats to patient safety. Beyond clinical harm, the resulting operational and financial fallout — including incorrect billing, legal liabilities, and malpractice claims — further underscores the urgency of developing robust defenses for MIoT systems (Gisselsson, 2022; Cohen et al., 2024). Despite these high-stakes risks, adversarial robustness for MIoT settings remains comparatively under-investigated relative to general IoT and conventional enterprise networks, motivating the domain-specific focus of this work.

Although the underlying reasons for adversarial attacks remain under active investigation, early studies suggested that these attacks arose due to rare pockets of vulnerability in the training process or because of the linearity in deep neural networks (Stutz et al., 2018). However, a widely recognized explanation now points to the complex geometry of data manifolds, the intrinsic structures where legitimate data is concentrated (Stutz et al., 2018). Foundational studies such as Gilmer et al. (2018) have revealed that adversarial attacks often lie in high-dimensional spaces that deviate from the true data manifold. Ma et al. (2018) further characterized these adversarial regions using local intrinsic dimensionality method, demonstrating that although these subspaces are near the data manifold, they occupy higher-dimensional areas where standard models typically struggle. Recent research has added nuance to this perspective by showing that

on-manifold adversarial examples also exist (Xiao et al., 2025). These inputs, residing within the data manifold, can still fool target models and sometimes even boost clean generalization, suggesting that the off-manifold assumption does not fully explain adversarial behavior. Although many defenses such as preprocessing, adversarial training, and adaptive model schemes have been proposed to detect adversarial attacks in cybersecurity, tabular data, and computer vision domains (see Section 3), existing methods often degrade accuracy on clean data, overlook manifold-based approaches, and fail to generalize across different attack types, leaving a critical gap that must be addressed.

Our defense strategy is motivated by two key observations: (1) adversaries often exploit access to input data, internal representations, or the model itself, using feature transferability to craft perturbations that fool the model; and (2) manifold-based defense strategies remain underexplored in the context of adversarial robustness for NIDS. To counter this, we intentionally make model representations more resistant to manipulation by establishing strong dependencies among features. As a result, attacks that target only a small subset of features are less likely to succeed, since altering one feature unpredictably affects others.

Specifically, our method projects network features into multiple distinct manifold spaces and intertwines these projections to establish robust interdependencies. We then apply cross-manifold mixup augmentation, which not only enriches the overall feature representation but also enforces greater consistency across manifold boundaries, thereby further diminishing the impact of adversarial attacks.

The contributions of this paper are as follows:

- We develop a novel manifold-based strategy (M4D)¹ that significantly enhances the robustness of DT models against adversarial attacks by leveraging multi-manifold feature representations.
- We validate our approach on two recent network MIoT datasets under multiple adversarial attack methods and benchmark it against five state-of-the-art defenses, demonstrating that our model is the only one to preserve high detection accuracy in both clean and adversarial scenarios across both datasets.

The remainder of this paper is organized as follows. Section 2 reviews relevant background, Section 3 discusses related work, and the threat model is outlined in Section 4. The proposed methodology is

¹ Code available at: <https://github.com/mohdah200/M4D.git>.

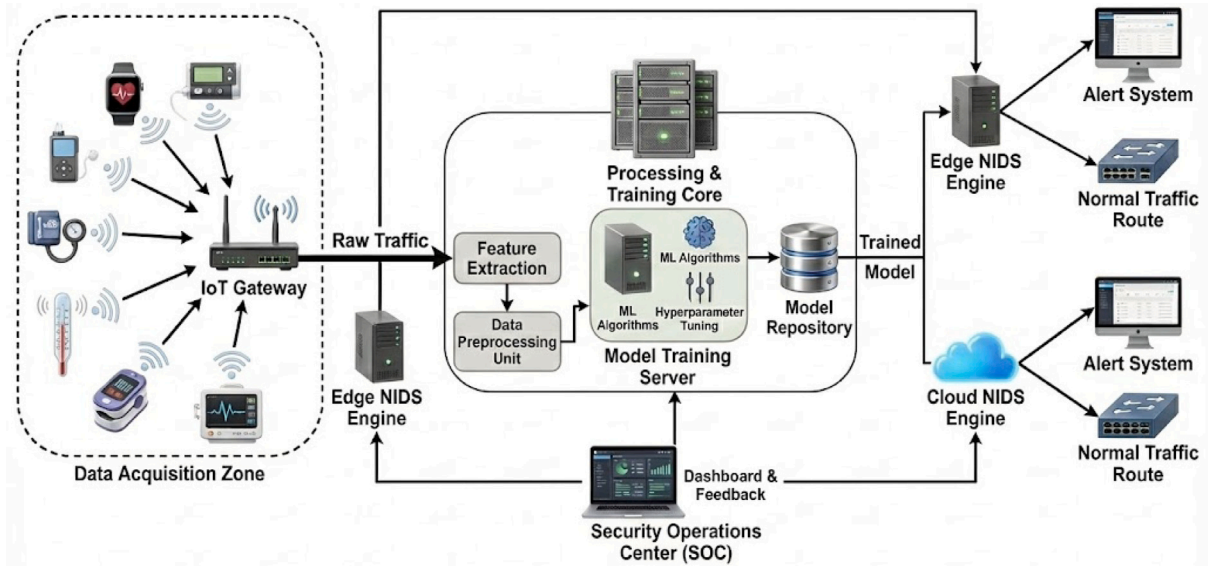


Fig. 2. The Design of ML-IDS.

presented in Section 5. Section 6 details the experimental setup, while Section 7 reports the results. Section 8 presents ablation studies, and Section 9 concludes the paper.

2. Background

2.1. Background on NIDS

NIDS are essential cybersecurity tools designed to continuously monitor network traffic and identify potential security threats in computer or IoT networks (Al-Fawa'reh et al., 2022). Traditionally, NIDS have relied on signature-based techniques, which cannot detect novel or zero-day attacks, require constant updates to remain effective, and can be easily evaded by slight modifications to known threats. To overcome these shortcomings, modern NIDS increasingly integrate ML algorithms to detect anomalies that deviate from normal behavior (Al-Fawa'reh et al., 2022). In this paper, we refer to the task of detecting anomalies in an environment without adversarial attacks as “clean detection”. The general process of ML-NIDS is summarized in Fig. 2, which illustrates the workflow that includes network traffic collection, feature extraction, training of the ML model, and the detection phase during inference.

2.2. Background on adversarial attacks

Adversarial attacks on ML models involve crafting small perturbations to input data to cause misclassifications (Vos and Verwer, 2023). Given an original input x with true label y and a classifier $f(\cdot)$, the adversary aims to find a perturbation δ such that the perturbed input $x' = x + \delta$ is misclassified by the model, while keeping δ small enough that x and $x + \delta$ are nearly indistinguishable under a norm constraint (e.g., $\|\delta\| \leq \epsilon$ for some small ϵ). The perturbation δ is often obtained by maximizing the model loss $\mathcal{L}(f(x'), y)$.

Fig. 1 illustrates a 2D view of how perturbations affect the data: malicious or normal samples are shifted, sometimes moving close to or even crossing the decision boundary of the opposite class. In the network traffic domain, the perturbation process is complex and must ensure that the traffic remains realistic. For example, when modifying features such as duration or packet size, these values must remain positive. Similarly, changes to network layer parameters (following TCP/IP standards) should avoid producing negative values or uniform flag settings, as such unrealistic alterations would be flagged as invalid

traffic and dropped by the IDS. In this paper, we address these challenges by incorporating these rules into the optimization process for generating adversarial perturbations.

3. Related work

Over the past few years, research on adversarial attacks has grown significantly on both offensive and defensive fronts, in what resembles a never-ending competition (Al-Fawa'reh et al., 2025). In this review of the literature, we focus on defensive methods. Specifically, we classify these techniques into two categories: Universal Defense (UD) methods, which are broadly applicable across various models, and DT Defense (DTD) methods, which are tailored specifically to DT models. Table 1 presents a comparison of several models, highlighting their advantages and limitations.

3.1. Universal defense

Defensive techniques applicable to any model generally focus on one of several strategies: preprocessing, data augmentation, adversarial retraining, or reducing the attack surface. One key preprocessing defense is the Denoising Autoencoder (DAE), which is trained to reconstruct clean inputs from deliberately corrupted examples by minimizing reconstruction error, thereby filtering out adversarial perturbations before classification (Sethi et al., 2020; Al-Hawawreh et al., 2021).

In terms of data augmentation, methods such as Gaussian Augmentation (GA) introduce random Gaussian noise during training to simulate diverse perturbations, thus improving the robustness of the model (Zantedeschi et al., 2017). Alternatively, strategies that reduce the attack surface include Feature Squeezing (FSQ) (Xu et al., 2018), which merges similar feature values to simplify inputs.

Adversarial training (Adv-Tr) represents another widely used defense mechanism, where the model is explicitly retrained using adversarial examples generated from one or more known attacks (Goodfellow et al., 2014). This process aims to enhance robustness by exposing the model to potential adversarial perturbations during training, effectively teaching it to recognize and resist specific forms of manipulation. In practice, Adv-Tr is often effective against attack types present in the adversarial training set, as the model learns to classify perturbed inputs correctly. However, its robustness may be limited when faced with novel or substantially different attack strategies that were not represented during training.

Table 1

Summary of related work on adversarial defenses for DT.

Model	Advantage	Limitation
DAE (Sethi et al., 2020; Al-Hawawreh et al., 2021)	Removes noise pre-classification	Limited to known perturbations
GA (Zantedeschi et al., 2017)	Simulates diverse noise	Ineffective for structured attacks
FSQ (Xu et al., 2018)	Simplifies inputs	Collapses under strong attacks
RAIDS (Sarikaya et al., 2023)	Filters off-manifold inputs	Limited generalization across diverse attacks
MAB (Paya et al., 2024)	Routes to least-vulnerable branch	Limited generalization across diverse attacks
PRB (Andriushchenko and Hein, 2019)	Provides tight certified bounds	Reduces accuracy on clean data
GROOT (Vos and Verwer, 2021a)	Adversarial-aware splits	Requires modifying the model
ROCT (Vos and Verwer, 2021b)	Yields optimal robust trees	Limited potential to improve clean accuracy
Genetic (Ranzato and Zanella, 2021)	Compact, interpretable robustness	Requires modifying the model
TREANT (Calzavara et al., 2019)	Evasion-aware loss-based splits	Limited potential to improve clean accuracy
FPRDT/PRAdaBoost (Guo et al., 2022)	Efficient training with guarantees	Requires modifying the model
TTTS (Cohen et al., 2024)	Test-time defense without retraining	Limited generalization across divers attacks
FairTTTS (Cohen-Inger et al., 2025)	Adds fairness adjustments post-hoc	Limited generalization across divers attacks

Recent defenses such as RAIDS (Sarikaya et al., 2023) and Apollon (Paya et al., 2024) present two distinct mechanisms for hardening NIDS against adversarial attacks. RAIDS computes reconstruction errors from autoencoders trained separately on benign and attack data, then fuses these signals with outputs from randomly partitioned ensemble classifiers in a LightGBM meta-classifier, achieving substantial robustness gains without any adversarial training (Sarikaya et al., 2023). Apollon, on the other hand, layers diverse classifiers behind a Thompson-sampling multi-armed bandit for adaptive model selection and employs request clustering to inject unpredictability, thwarting adversarial ML attacks without requiring training on adversarial attacks (Paya et al., 2024). In the remainder of the paper, we refer to this method as MAB.

3.2. Decision tree defense

Recent work on adversarial robustness in DTs explores a spectrum of trade-offs between clean accuracy, computational efficiency, and scalability. Andriushchenko and Hein (2019) propose Provably Robust Boosted (PRB), a boosting-based framework that computes the exact robust loss for decision stumps and derives a tight, efficiently optimizable upper bound for deeper trees. GROOT (Vos and Verwer, 2021a) shifts the focus to efficiency by replacing the standard Gini impurity with an analytically derived adversarial variant, enabling split evaluation under worst-case perturbations in $O(n \log n)$ time but relying on greedy decisions that may sacrifice global optimality. ROCT (Vos and Verwer, 2021b) takes the opposite extreme, formulating adversarial training under the 0–1 loss as a single mixed-integer optimization problem solved; this yields globally optimal robust trees at the cost of scalability on deep or large datasets. Together, these methods map out the design space for DT robustness: from exact, provable bounds, through fast heuristics to optimization-based exact solutions, each balancing robustness, tractability, and scalability in a different way. Evolutionary strategies have been explored as well. Ranzato and Zanella (2021) employ a genetic algorithm augmented with formal verification via abstract interpretation to evolve robust DT ensembles. While this approach often produces compact, interpretable, and robust models, it is computationally intensive and sensitive to hyperparameter settings. In contrast, TREANT (Calzavara et al., 2019) integrates robust splitting and attack invariance into a greedy tree-construction process that minimizes an evasion-aware loss function, yielding significant improvements in ROC AUC at the expense of increased computational cost. Guo et al. (2022) introduce Fast Provably Robust Decision Trees (FPRDT) and its boosting extension PRAdaBoost, which directly minimizes the adversarial 0/1 loss while maintaining $O(n \log n)$ complexity and offering exponential convergence guarantees—achieving provable robustness more efficiently than prior methods.

Cohen et al. (2024) argue that defenses such as those in Andriushchenko and Hein (2019), Vos and Verwer (2021a,b), Ranzato and Zanella (2021), Calzavara et al. (2019) and Guo et al. (2022) incur a degradation in clean accuracy. Cohen et al. (2024) proposed TTTS

(Tree Test Time Simulation), an inference-time technique designed to boost the adversarial robustness of DTs without modifying their underlying structure or retraining the model. Instead of altering the tree or its parameters, TTTS introduces a Monte Carlo simulation during test time. For each test instance, the algorithm follows the tree's standard decision path but with a probability determined by one of several heuristics it occasionally deviates to an alternative branch. The final prediction is computed by averaging the outcomes over multiple simulated paths. Building on TTTS, the authors of Cohen-Inger et al. (2025) introduced FairTTTS (FTTTS), which aims to reduce bias while enhancing predictive accuracy. FTTTS employs a distance-based heuristic to adjust decisions at protected attribute nodes, thereby promoting fairness for underrepresented samples. This fairness-oriented adjustment is performed as a post-processing step, making it adaptable for use with pre-trained models. A prevalent limitation among defenses in the DTD category is a compromise in clean-data accuracy relative to non-robust models, a trade-off highlighted by Cohen et al. (2024). While approaches such as TTTS and FTTTS are notable exceptions that preserve high clean accuracy, our evaluation in Section 7 reveals that they fail to generalize, suffering significant performance degradation under adversarial attack. Consequently, there is a critical need for a defense strategy that simultaneously maintains high clean-data performance and ensures robustness against a diverse range of adversarial perturbations.

4. Threat model

This work investigates the robustness and generalizability of decision tree-based NIDS deployed in IIoT environments under adversarial attack. We focus on realistic scenarios where different healthcare organizations may implement similar anomaly detection solutions, yet use distinct feature extraction engines and cybersecurity practices.

From an architectural standpoint, the attack surface in IIoT mirrors traditional IoT and computer networks (e.g., compromised gateways or edge nodes). However, our study prioritizes the unique and less-explored characteristics of IIoT traffic, which play a crucial role in shaping attack strategies. Unlike the bursty, user-driven traffic typical of consumer IoT, IIoT devices generate continuous, periodic physiological data streams (e.g., heart rates, insulin levels) that exhibit high stationarity. This regularity creates distinct feature manifolds that our defense specifically exploits. Attackers in this domain face a unique constraint: they must manipulate these predictable streams to evade detection without violating the strict validity ranges inherent to medical protocols.

As illustrated in Fig. 3, our setting considers two independent organizations (Organization A and Organization B), each operating a DT-based NIDS with a unique feature extraction pipeline. We examine both white-box and black-box adversarial scenarios acting upon these victims. In the white-box case, depicted by the blue “Full Knowledge” path, the attacker possesses complete access to the specific organization's model and feature extraction process. In the black-box case, the attacker operates a surrogate system (e.g., targeting Organization A)

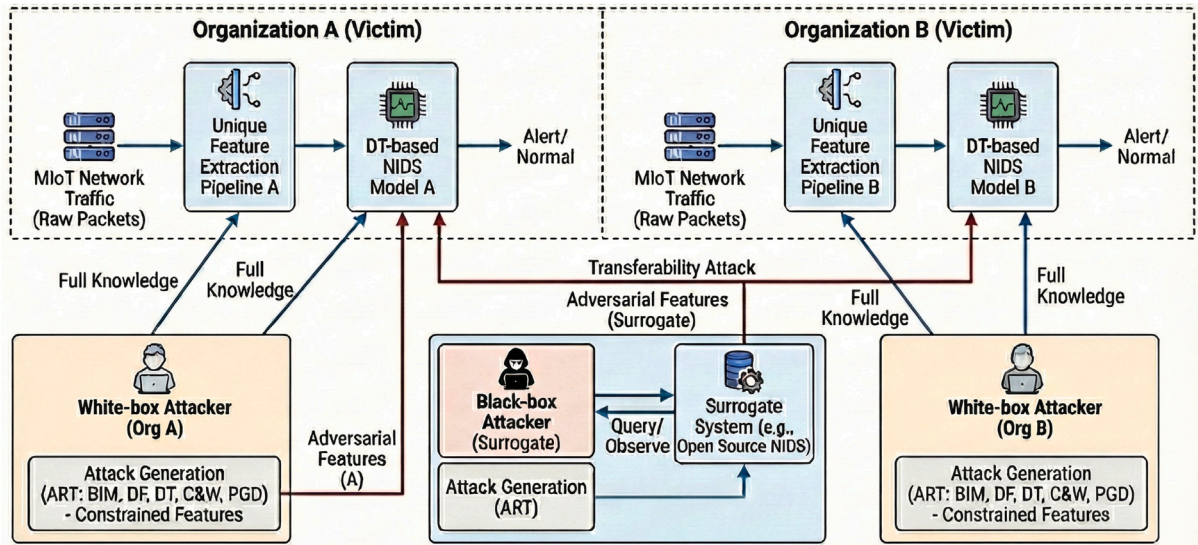


Fig. 3. The proposed adversarial evaluation framework for MIoT. The model considers heterogeneity in feature extraction pipelines between Organization A and Organization B to assess defense robustness against both direct (Full Knowledge) and transferred adversarial attacks.

and relies on the transferability of adversarial examples to bypass the defenses of a different target (e.g., Organization B). This simulates common industry situations where threat actors leverage knowledge from widely used or partially open-source detection tools to compromise distinct proprietary systems (Papernot et al., 2016).

Under this threat model, we additionally benchmark a set of established adversarial methods that have been shown effective in other IoT settings, conventional computer networks, and tabular learning problems. However, our empirical analysis reveals that these off-the-shelf attacks and defenses do not consistently translate to the MIoT DT-based NIDS setting: several methods exhibit noticeably reduced effectiveness and poor transferability when confronted with heterogeneous feature pipelines and domain-specific constraints. This gap highlights that robustness claims established in broader IoT/tabular contexts may not hold in MIoT deployments, reinforcing the need for defenses explicitly designed and validated for this domain.

5. Proposed method

In this section, we introduce M4D, a manifold-based defense framework for DT-based NIDS. As shown in Fig. 4, M4D's preprocessing pipeline comprises manifold extraction and projection, manifold entanglement, cross-manifold mixup (applicable to clean samples alone or to both clean and adversarial samples generated by any existing algorithm, such as FGSM (Goodfellow et al., 2014)), and manifold regularization. Together, these stages transform raw inputs into richer, interdependent representations that bolster the robustness and generalizability of downstream tree-based models.

Existing defense methods often fail to guarantee generalizable robustness across diverse attack types and varying data dimensionalities. M4D addresses this by prioritizing Manifold Entanglement to enforce strict interdependencies between feature subspaces. While attention mechanisms and mixup are established techniques, prior works have not utilized them to enforce such entanglement. Our ablation study demonstrates that these components individually cannot guarantee robustness; however, applying the entanglement concept significantly increases the difficulty for the attacker while maintaining stable clean accuracy. To the best of our knowledge, this is the first study to explore latent space entanglement within the NIDS domain, and our experiments confirm that this specific integration is critical for achieving effective defense where other methods fail.

5.1. Manifold extraction and projection

M4D begins by extracting manifolds from the input sample. Let $\mathbf{x} \in \mathbb{R}^d$ be an input sample, which is processed by the encoder $\mathcal{E}$ to yield a hidden representation $\mathbf{h} = \mathcal{E}(\mathbf{x}) \in \mathbb{R}^H$, where H is the dimensionality of the hidden representation. Subsequently, we apply a well-established projection head $\mathcal{P}_m$ (Xue et al., 2024) for each manifold m (with $m = 1, \dots, M$) to map $\mathbf{h}$ into a specific manifold space, such that $\mathbf{z}_m = \mathcal{P}_m(\mathbf{h})$ with $\mathbf{z}_m \in \mathbb{R}^{D_m}$. This manifold extraction and projection step yields a set of manifold projections $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_M\}$, each capturing distinct feature aspects or “views” of the data.

Projection heads are employed because they decouple the general features learned by $\mathcal{E}$ from the more specialized, manifold-specific representations. By mapping $\mathbf{h}$ into distinct manifold spaces via $\mathcal{P}_m$, each projection head focuses on capturing unique aspects of the data. This specialization is crucial for subsequent processes, such as manifold entanglement, cross-manifold mixup, and regularization, and ultimately leads to a more robust and resilient model, particularly under adversarial conditions.

5.2. Manifold entanglement

The idea behind manifold entanglement is to fuse separate feature manifolds into a single interdependent representation so that perturbations in one manifold immediately diffuse across all others. This is especially useful against low-budget attacks, where minimal, stealthy changes must now distort multiple manifolds, making detection easier. This entanglement can be achieved through cryptographic operations, deterministic fusion rules (e.g., constrained weighted averaging), and learnable fusion modules. We focus on learnable methods because they can automatically adapt to the data distribution and capture complex, non-linear interdependencies across manifolds through training. To this end, we introduce entanglement via multi-head attention (Vaswani et al., 2017) (denoted as $\mathcal{A}$). For a given manifold m , we stack the individual manifold projections into a unified representation matrix $\mathbf{Z} \in \mathbb{R}^{M \times D}$, Where D is the (assumed uniform) dimensionality for simplicity. The multi-head attention mechanism $\mathcal{A}$ is then applied to $\mathbf{Z}$ to enable each manifold to incorporate information from all others. In standard multi-head attention, for a given head h the attention is computed as in Eq. (1).

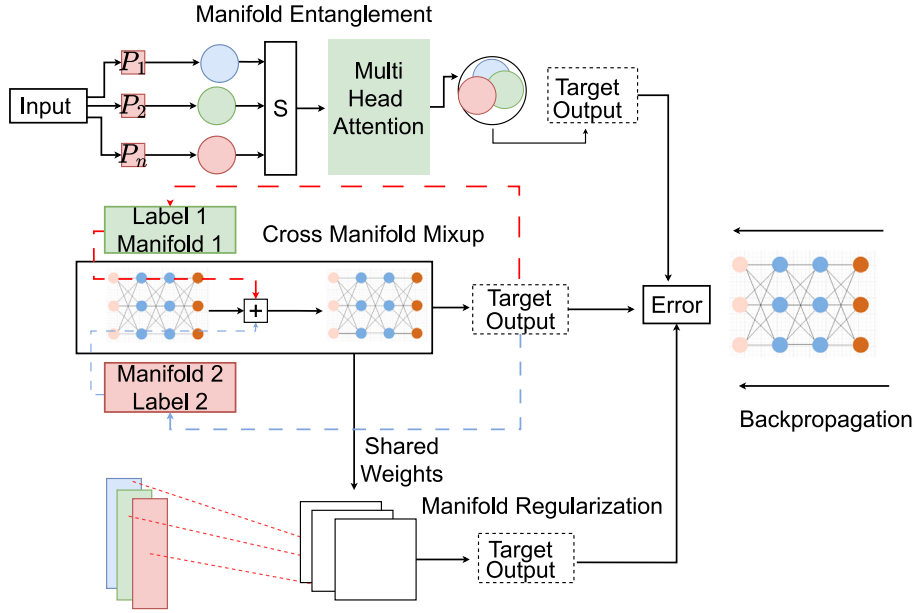


Fig. 4. Overview of M4D Framework.

$$\text{head}_h = \text{softmax}\left(\frac{(\mathbf{Z}\mathbf{W}_h^Q)(\mathbf{Z}\mathbf{W}_h^K)^T}{\sqrt{d_k}}\right)(\mathbf{Z}\mathbf{W}_h^V), \quad (1)$$

where $\mathbf{W}_h^Q, \mathbf{W}_h^K, \mathbf{W}_h^V \in \mathbb{R}^{D \times d_k}$. The outputs from all heads are then concatenated and projected as in Eq. (2).

$$\mathcal{A}(\mathbf{Z}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)\mathbf{W}^O \quad (2)$$

With $\mathbf{W}^O \in \mathbb{R}^{(H \cdot d_k) \times D}$.

As in Eq. (3), for each manifold m , we extract its updated representation by selecting the m th element from the attention output, and then refine it with a residual connection and layer normalization (denoted as $\mathcal{N}(\cdot)$).

$$\mathbf{z}'_m = \mathcal{N}\left(\mathbf{z}_m + [\mathcal{A}(\mathbf{Z})]_m\right) \quad (3)$$

This intermediate representation is further processed by a feed-forward network $\mathcal{F}(\cdot)$ (typically a two-layer network with a ReLU activation), again with a residual connection and layer normalization (see Eq. (4)).

$$\mathbf{z}''_m = \mathcal{N}\left(\mathbf{z}'_m + \mathcal{F}(\mathbf{z}'_m)\right) \quad (4)$$

Here, $\mathbf{z}''_m$ denotes the final entangled projection for manifold m . This manifold entanglement mechanism is designed to fuse complementary information across different manifold representations. By enabling each manifold projection to attend to and incorporate cues from the others, we achieve richer and more robust feature representations. The residual connections ensure that the original manifold-specific information is preserved while still benefiting from the cross-manifold interactions, and the layer normalization $\mathcal{N}(\cdot)$ stabilizes the training process by normalizing these updates. Consequently, the resulting set of enriched representations $\{\mathbf{z}''_1, \dots, \mathbf{z}''_M\}$ becomes more resilient to adversarial perturbations, as any attack targeting a single manifold is mitigated by the aggregated information from multiple manifolds.

5.3. Cross-manifold mixup augmentation

To further smooth decision boundaries and cover “blind spots” in the latent space, we introduce *Cross-Manifold Mixup*. Unlike standard mixup which interpolates within the same feature space, our strategy enforces consistency between different manifolds (Verma et al., 2018).

For a sample i with entangled projection $\mathbf{z}''_{m,i}$, we select a pairing manifold $m' = (m \bmod M) + 1$ from a random sample j . We then generate a mixed feature vector $\tilde{\mathbf{z}}_m$ and a mixed label $\tilde{y}$ by linearly interpolating between manifold m of sample i and manifold m' of sample j using a mixing coefficient $\lambda \sim \text{Beta}(\alpha, \alpha)$. This forces the model to learn robust features that are invariant to the specific manifold source, significantly enhancing generalization. The final mixed features from all manifolds are concatenated and passed to the classifier.

5.4. Training and regularization

M4D optimizes a joint loss function designed to ensure classification accuracy while enforcing structural consistency in the manifold space. We introduce a *manifold regularization* term, $\mathcal{L}_{\text{reg}}$, which penalizes discrepancies between the clean entangled projections $\mathbf{z}''_m$ and their mixup-augmented counterparts $\tilde{\mathbf{z}}_m$:

$$\mathcal{L}_{\text{reg}} = \frac{1}{M} \sum_{m=1}^M \|\mathbf{z}''_m - \tilde{\mathbf{z}}_m\|_2^2 \quad (5)$$

The total training objective is a weighted sum of the clean cross-entropy loss ($\mathcal{L}_{\text{CE}}$), the mixup loss ($\mathcal{L}_{\text{Mixup}}$), the manifold regularization, and an L_1 sparsity penalty:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) \mathcal{L}_{\text{Mixup}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \mathcal{L}_{L1} \quad (6)$$

Here, α balances clean and robust accuracy, while λ_{reg} controls the strength of the manifold consistency constraint.

6. Experiments

For evaluation, we first assess the traditional intrusion detection task under clean conditions, using normal network traffic and malicious attack traffic, including attacks such as DoS and data injection. We then evaluate performance under five distinct adversarial attack scenarios. We compare M4D against six other defenses: FSQ (Xu et al., 2018), Adv-Tr (Papernot et al., 2016), RAIDS (Sarikaya et al., 2023), MAB (Paya et al., 2024), TTTS (Cohen et al., 2024), and FTTTS (Cohen-Inger et al., 2025), using classification accuracy, which is appropriate given our balanced training and testing sets. FSQ is a simple and easy-to-implement defense that can be readily applied across various domains, including both computer vision and tabular data tasks. Adv-Tr serves

Table 2

Number of samples in train and test sets.

	CICIoMT2024	ICU
Train/Test	385 464/75 214	128 198/32 050

Table 3

Training and testing time comparison.

	M4D	DT
Tr (s)/Ts (s)	612.40/0.92	69.41/0.11

as a widely adopted baseline employing adversarial training to enhance robustness. RAIDS and MAB are recent methods that have demonstrated robustness against certain adversarial attacks in the literature, specifically within the NIDS domain; here, we further evaluate their generalizability across a wider range of attack scenarios. TTTS and FTTTS are post-hoc methods applied at inference time. TTTS has been shown to significantly improve the accuracy of decision trees across diverse datasets, as demonstrated in [Cohen et al. \(2024\)](#), while FTTTS is an enhanced version of TTTS, incorporating additional mechanisms for improved performance. This diverse selection of defenses enables a comprehensive evaluation of M4D relative to both established baselines and state-of-the-art methods. For M4D, we also report AUC-ROC scores and present confusion matrices for both datasets.

6.1. Experimental setup

M4D uses two 256-dimensional manifold projections, an 8-head attention mechanism for entanglement, and 512-unit hidden layers in both the encoder and the classifier. Training was conducted with the Adam optimizer (learning rate 1×10^{-3} , $(\beta_1, \beta_2) = (0.9, 0.999)$), a mini-batch size of 128, and exactly 20 epochs per fold. After each epoch, we evaluated on a held-out 20% of that fold's data and saved the parameters from the epoch achieving the highest validation accuracy. Manifold mixup was applied by empirically sampling $\lambda \sim \text{Beta}(\alpha = 0.4, \alpha = 0.4)$ and clamping $\lambda \geq 0.5$, with cross-manifold mixing enabled. We incorporated manifold regularization, which measures the mean-squared error between clean and mixed projections, with weight $\lambda_{\text{reg}} = 0.1$, and applied L1 regularization on all model parameters with $\lambda_{\text{L1}} = 1 \times 10^{-5}$. We performed three-fold cross-validation (shuffle with fixed seed) on the training set to select the best snapshot across all folds. Finally, we extracted the concatenated manifold features from the best model on the full train/test splits and trained a DT model on these representations for our baseline evaluation. We performed a three-fold cross-validation on the training dataset, allocating 20% of each fold's data for validation. Our total training loss is a weighted sum of cross-entropy for clean data, a mixup-based loss (computed via KL divergence with the mixed labels), manifold regularization, and L1. This combined loss function is minimized via gradient descent for a maximum of $N = 2000$ iterations per fold (or fewer if early stopping triggers). All experiments were conducted on a machine with an AMD Ryzen Threadripper 3970X (64 cores, 4.55 GHz max), 125 GiB RAM, a 916 GiB NVMe drive, and an NVIDIA GeForce RTX 3060 (12 GB VRAM).

6.2. Dataset

Although our method is broadly applicable to various anomaly detection and network security scenarios, we evaluated it using two recent and realistic Medical IoT (MIoT) traffic benchmarks. This choice reflects the urgent need for robust anomaly detection in healthcare settings, where the consequences of misclassification can be severe.

In MIoT and healthcare systems, interconnected devices are now integral to patient monitoring, diagnostics, and automated interventions. Inaccurate anomaly detection can have significant impacts: a missed threat may result in delayed or inappropriate treatment, while false alarms can lead to alarm fatigue, unnecessary procedures, and

Table 4

Performance of M4D and baselines under adversarial attacks on CICIoMT2024.

Bold = best, **italic bold** = second best.

Case	FSQ	Adv-Tr	RAIDS	MAB	TTTS	FTTTS	M4D
Clean	0.5000	0.9983	0.9710	0.9491	0.9984	0.9988	0.9971
BIM	0.5000	0.9929	0.4855	0.9358	0.9863	0.9919	0.9173
DF	0.5000	0.5962	0.4855	0.9494	0.6108	0.9910	0.9223
DT	0.5000	0.9989	0.4855	0.9484	0.5177	0.5189	0.9072
C&W	0.5000	0.8426	0.4855	0.8169	0.8433	0.8545	0.9044
PGD	0.5000	0.9957	0.4855	0.9309	0.5110	0.8215	0.9171
Avg	0.5000	0.9041	0.5664	0.9218	0.7446	0.8628	0.9276

increased operational costs. These risks, combined with potential legal and financial repercussions, make detection accuracy a matter of both safety and economic necessity.

The first dataset, CICIoMT2024 ([Dadkhah et al., 2024](#)), released in 2024, captures network traffic from a 40 device MIoT testbed using WiFi, MQTT, and Bluetooth protocols. It includes both benign and malicious traffic across five attack types: DDoS, DoS, reconnaissance, MQTT floods, and malformed data. Detailed profiling of device behavior across different operational states ensures high realism.

The second dataset, ICU Healthcare Security ([Hussain et al., 2021b,a](#)), published in 2021, was collected from an intensive care simulation with two beds, each equipped with patient monitoring sensors and an environmental controller. It covers both normal and adversarial scenarios, including MQTT publish floods, MQTT authentication bypass, packet crafting, and CoAP replay attacks, using real world medical protocols.

These datasets were chosen for their protocol diversity, device heterogeneity, and clinical relevance, qualities often lacking in generic traffic datasets. Our approach is not limited to these datasets and can be extended to other domains and network environments. [Table 2](#) summarizes the network flow samples used. Further details on feature extraction and data collection can be found in the original dataset publications ([Dadkhah et al., 2024](#); [Hussain et al., 2021b,a](#)).

6.3. Feature selection and preprocessing

For ICU dataset ([Hussain et al., 2021b,a](#)), we retained the complete set of network- and application-layer features extracted from each generated pcap file, while for CICIoMT2024 ([Dadkhah et al., 2024](#)) we restricted our inputs to statistical flow features only. This pared-down selection yields comparable detection accuracy to the full feature set but with lower computational cost, making it well suited for resource-constrained environments. In both cases, we first removed any records containing missing values, then eliminated duplicate network-flow entries to prevent skewed representation, and finally applied feature normalization to scale all values into a common range and avoid bias during model training.

6.4. Computational resources

[Table 3](#) summarizes the timing results. Training on 385 464 samples required 612.40 s for the manifold-defense framework and 69.41 s for the DT classifier. Inference on 75 214 test samples completed in just 0.11 s. These results show potential for edge deployment, with training typically performed offline to eliminate the need for continuous retraining.

7. Results and discussion

7.1. Results

[Table 4](#) reports classification accuracy on the CICIoMT2024 dataset for seven defenses under clean and adversarial scenarios. Under the clean scenario, FSQ yields 50.00%, Adv-Tr 99.83%, RAIDS 97.10%, MAB 94.91%, TTTS 99.84%, FTTTS 99.88%, and M4D 99.71%. When

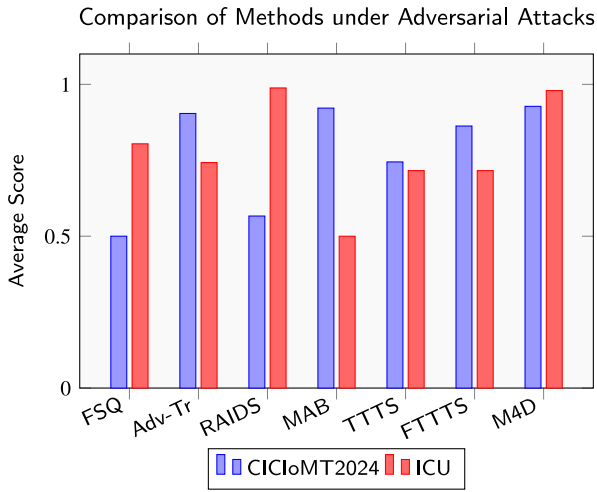


Fig. 5. Comparison of average method performance under adversarial attacks on CICIOMT2024 and ICU datasets.

subjected to BIM, DF, DT, C&W, and PGD attacks, FSQ remains at chance level (50.00%) and RAIDS drops to an average of 56.64%. Adv-Tr performs well in most cases but exhibits notable drops under DF (59.62%) and C&W (84.26%), resulting in an overall average of 90.41%. MAB maintains accuracy above 90% for most attacks except C&W (81.69%). TTTS shows reduced performance under DF, DT, and PGD (ranging from 51.10% to 61.08%). FTTTS performs well in all cases except DT (51.89%). M4D consistently maintains accuracies between 90.44% and 92.23% across all adversarial attacks. Averaging across all scenarios, M4D achieves the highest mean accuracy (92.76%), followed by MAB (92.18%), Adv-Tr (90.41%), FTTTS (86.28%), TTTS (74.46%), RAIDS (56.64%), and FSQ (50.00%). Table 5 presents the same comparison on the ICU dataset. Under the clean scenario, FSQ, Adv-Tr, RAIDS, TTTS, FTTTS, and M4D all exceed 99.04%, with Adv-Tr, RAIDS, TTTS, FTTTS, and M4D at 100%. Under BIM, DF, DT, C&W, and PGD attacks, M4D sustains high accuracy (93.96 to 100%, average 97.96%). FSQ drops (62.76 to 99.04%, average 80.42%), Adv-Tr shows more variability (52.01 to 100%, average 74.25%), MAB remains at 50.00%, and TTTS/FTTTS drop to 50.00 to 63.69% (average 71.60%). In contrast, RAIDS experiences only a minor reduction relative to its clean accuracy (97.57 to 100%, average 98.82%). Fig. 5 presents a comparison of the average accuracy across all scenarios and models for both datasets. Notably, most models exhibit distinct behaviors under different conditions (see Section 7.2 for further details).

Table 6 shows that, under clean conditions, M4D achieves nearly perfect separation on both CICIOMT2024 (37,600 TN, 37,399 TP, 7 FP, 208 FN) and ICU (16,025 TN, 16,025 TP, 0 FP, 0 FN). When adversarial attacks are introduced, false positives surge on CICIOMT2024, rising to 5846, while false negatives remain low, peaking at 1346 under C&W. Different attacks affect true positives unevenly: DF leaves zero false negatives on CICIOMT2024, whereas BIM and PGD incur only modest drops. On ICU, only BIM and PGD create false negatives (up to 1935), while DT and C&W produce no errors.

As outlined in Table 6, on the CICIOMT2024 dataset, M4D incurs a higher rate of false-positive alarms under attack but maintains a low rate of false negatives. By prioritizing the reduction of false negatives, we accept an increase in false positives to ensure minimal missed detections. Although elevated false positives can lead to alert fatigue and increased operational overhead, this trade-off is generally acceptable in critical infrastructure or medical device settings, where missing a true threat, especially a zero-day exploit with no prior signature, could have catastrophic consequences. Practitioners can still adjust detection

Table 5

Performance of M4D and baselines under adversarial attacks on ICU. Bold = best, **bold italic** = second best.

Case	FSQ	Adv-Tr	RAIDS	MAB	TTTS	FTTTS	M4D
Clean	0.9904	1.0000	1.0000	0.5000	1.0000	1.0000	1.0000
BIM	0.6276	0.5201	0.9757	0.5000	0.6340	0.6340	0.9396
DF	0.6365	0.5258	0.9780	0.5000	0.6369	0.6369	0.9984
DT	0.9904	1.0000	1.0000	0.5000	0.5000	0.5000	1.0000
C&W	0.9525	0.8890	1.0000	0.5000	0.8912	0.8912	1.0000
PGD	0.6276	0.5201	0.9757	0.5000	0.6340	0.6340	0.9396
Avg	0.8042	0.7425	0.9882	0.5000	0.7160	0.7160	0.9796

Table 6

Confusion matrix results for CICIOMT2024 and ICU datasets.

Case	CICIOMT2024				ICU			
	TN	FP	FN	TP	TN	FP	FN	TP
Clean	37 600	7	208	37 399	16 025	0	0	16 025
BIM	31 761	5846	377	37 230	16 025	0	1935	14 090
DF	31 761	5846	0	37 607	16 025	0	52	15 973
DT	31 761	5846	1136	36 471	16 025	0	0	16 025
C&W	31 761	5846	1346	36 261	16 025	0	0	16 025
PGD	31 761	5846	388	37 219	16 025	0	1935	14 090

thresholds to rebalance specificity and sensitivity if the volume of false positives becomes burdensome.

7.2. Discussion

Our evaluation demonstrates that M4D provides a robust and generalizable defense across datasets with varying feature complexity. On CICIOMT2024 (Table 4), which has a smaller set of features, M4D achieves the highest average accuracy (92.76%) across all scenarios. In contrast, RAIDS, while leading on the more feature-rich ICU dataset (98.82% average), suffers a pronounced drop on CICIOMT2024 (56.64%). This contrast illustrates that defenses performing well in high-dimensional or feature-rich environments may not generalize effectively when fewer or less informative features are available.

Other baselines show similar limitations. TTTS and FTTTS deliver high clean accuracy on both datasets but are highly susceptible to adversarial attacks, especially on ICU, where their accuracy drops to the 50%–63% range. FSQ does not guarantee high clean accuracy, particularly in low-dimensional settings; on CICIOMT2024, FSQ achieves only 50.00% accuracy even in the absence of adversarial attacks, while in feature-rich environments like ICU, FSQ's clean accuracy exceeds 99%. This underscores the importance of considering feature set size in both robustness and nominal performance.

MAB also demonstrates a strong dependence on feature space. While it achieves high accuracy across most adversarial cases on CICIOMT2024, its accuracy is limited to 50.00% on the ICU dataset, irrespective of scenario. This highlights that MAB's modeling approach does not transfer well between different feature regimes.

Adv-Tr shows that adversarial training can be effective when the nature of the attack is known in advance. For example, the model performs very well against DT attacks, since training samples from the same attack are included during adversarial training. However, its robustness is less consistent against attacks that differ from those used in training, such as DF and C&W. On CICIOMT2024, Adv-Tr achieves high accuracy against attacks seen during training but experiences significant drops against others, reflecting the challenge of anticipating all possible attack strategies in a real-world deployment. Additionally, its performance exhibits greater variability on the ICU dataset, again emphasizing that adversarial training is most robust when the threat model matches the deployed environment.

M4D's consistent strength arises from its integration of manifold entanglement and cross-manifold mixup (see Section 8 for the impact

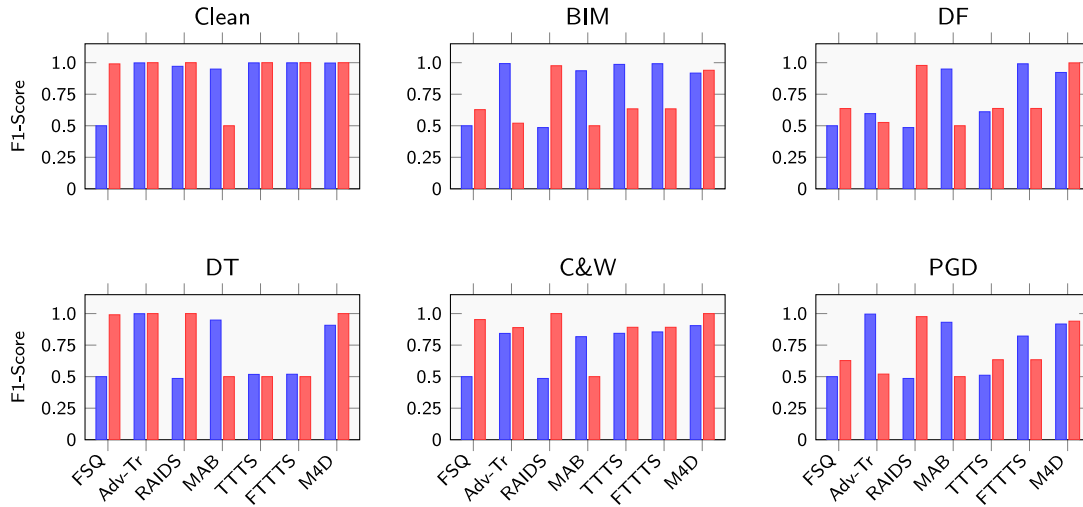


Fig. 6. F1-score comparison of defense methods across attack conditions on CICIoMT2024 and ICU datasets.

of each component on the overall design). Manifold entanglement enforces dependencies among features, increasing the complexity of adversarial perturbations, while cross-manifold mixup smooths decision boundaries and reduces vulnerability to adversarial “blind spots”. For example, in CICIoMT2024, the transformation of 16 features into a 512-dimensional latent space substantially raises the bar for successful attacks. Embedding visualizations (see Fig. 8) confirm that M4D maintains a clear separation between normal and adversarial samples, regardless of feature count or attack type.

Our findings demonstrate that high clean accuracy in a high-dimensional, feature-rich space (such as the ICU dataset) does not guarantee robust generalization, especially in lower-dimensional settings like the CICIoMT2024 dataset; conversely, the opposite also holds true, where even baseline methods such as FSQ and MAB may fail to deliver reliable detection. Only defenses like M4D consistently maintain high performance under both clean and adversarial conditions and across different feature regimes (see Figs. 5 to 7). This balance of robustness and adaptability is essential for practical NIDS deployment in diverse and evolving IIoT environments.

In this paper, we study adversarial attacks in the NIDS context to understand how carefully crafted perturbations can undermine even state-of-the-art intrusion detectors. Early evaluations of NIDS focused almost exclusively on clean-traffic accuracy, but adversarial testing has revealed that high nominal performance often masks critical vulnerabilities. By framing NIDS evaluation as an adversarial game, we expose hidden weaknesses in model decision boundaries and subtle feature dependencies that attackers can exploit to evade detection. This shift in perspective has driven the community to move beyond one-off accuracy benchmarks and toward worst-case robustness metrics, spurring the development of adversarial training, ensemble strategies, and input-space regularization techniques tailored for network traffic anomaly detection.

Finally, although this study prioritizes NIDS for IIoT due to the severe safety implications of missed detections in healthcare infrastructure, the core principles of M4D are largely classifier agnostic. In particular, manifold entanglement and cross-manifold mixup are designed to harden decision boundaries for tabular classifiers in domains susceptible to evasion attacks. Consequently, the framework can be adapted to related settings, such as conventional computer-network NIDS built on CIC-style traffic datasets where features are extracted using CICFlowMeter, a pipeline widely adopted across benchmark datasets in the literature and also used to construct the IIoT features in this work. Beyond network security, M4D can be extended to other security-critical applications, including financial fraud detection (see Section 8.3) and industrial control system monitoring. Such

Table 7

Ablation of M4D components on CICIoMT2024.

Scenario	Clean	BIM	DF	DT	C&W	PGD
All	0.9971	0.9173	0.9223	0.9072	0.9044	0.9171
Ent	0.9973	0.7224	0.7225	0.7185	0.7185	0.7223
Mix	0.9973	0.8989	0.4678	0.9026	0.8763	0.8944

adaptations would primarily require aligning the projection heads and preprocessing with the target feature dimensionality and data distribution (and re-tuning hyperparameters), while keeping the core defense mechanisms unchanged.

8. Ablation study

8.1. Ablation study of M4D component

To evaluate the contribution of each M4D component, we conducted an ablation study in which individual components were selectively disabled. We compared three configurations: the full model (All), which incorporates both manifold entanglement and multi-manifold mixup; a configuration employing only manifold entanglement (Ent); and one using only multi-manifold mixup (Mix). Table 7 reports the performance of each configuration under clean conditions and against various adversarial attacks (BIM, DF, DT, C&W, PGD). “All” achieves high accuracy on both clean data (99.71%) and under adversarial attacks (over 90% for every attack), underscoring the strength of integrating both components. “Ent” yields a slight increase in clean accuracy (99.73%) but sees substantial adversarial performance drop, with accuracies averaging around 72% across attacks. “Mix” maintains robust performance against BIM, DT, C&W, and PGD with accuracies close to “All”; however, it is notably more vulnerable to DF, where accuracy falls sharply to 46.78%. These results indicate that while each component contributes to adversarial robustness, their combined use in the full model is critical for achieving optimal defense. “Ent” provides a baseline defense, lifting adversarial accuracy to around 72%, whereas “Mix” substantially improves resistance to most perturbations; together, they form a complementary strategy that enhances the overall resilience of our model.

8.2. Ablation study of attacker knowledge

In this ablation study, we assess how varying levels of attacker knowledge affect our defense by considering two grey-box scenarios (Proj-only and Ent-only) and a fully adaptive scenario. In all cases, the

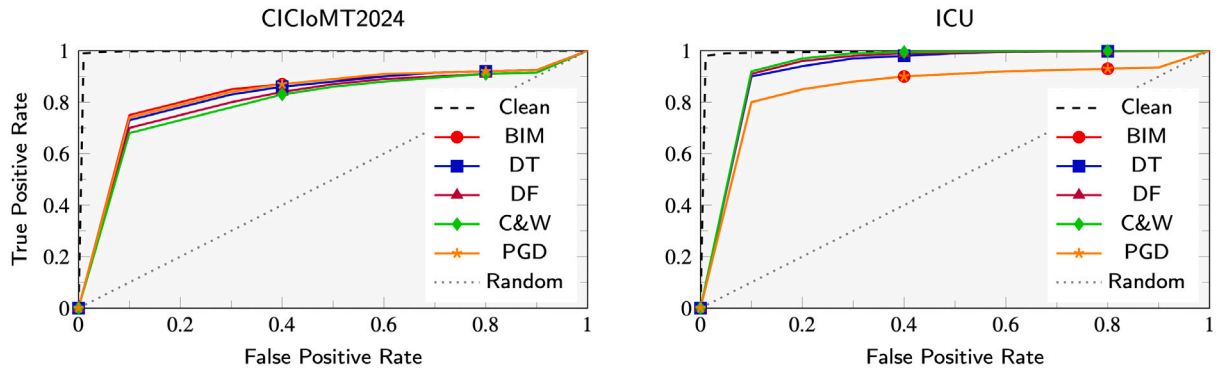


Fig. 7. ROC Curves for CICIoMT2024 and ICU Datasets.

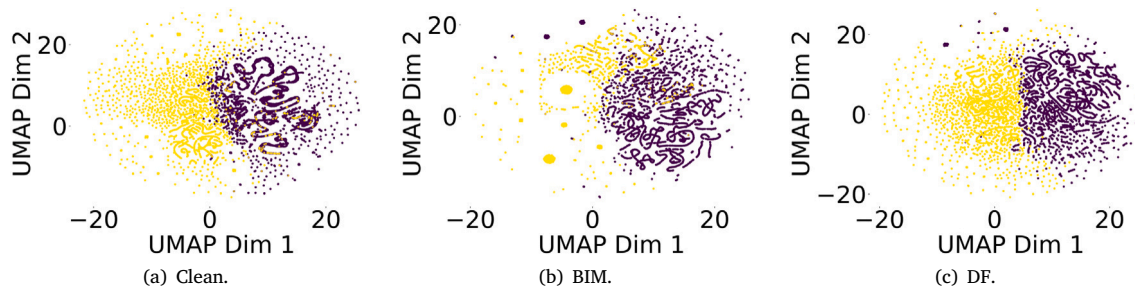


Fig. 8. Classification embeddings of normal and adversarial samples using the M4D defense on the CICIoMT2024 dataset, where yellow represents attack samples and purple represents normal samples.

attacker also has access to the DT model. When the attacker observes only the projection outputs before entanglement (Proj-only), accuracy drops marginally from 99.71% to 99.35%. Granting access to the entangled manifold features (Ent-only) yields a more noticeable decline to approximately 90.38% (see Table 8), yet the system retains strong robustness. Finally, under full adaptive knowledge (where the attacker knows the raw inputs, neural encoder, entanglement module, and DT), accuracy plummets to 53.04%, demonstrating that only comprehensive exposure can significantly compromise the defense. We recommend deploying this defense in a black-box or grey-box configuration without disclosing its internal components, in line with enterprise cybersecurity best practices.

8.3. Ablation study: Financial fraud generalization

To validate the domain-agnostic capabilities of M4D, we extended our evaluation to a Credit Card Fraud dataset (Cohen et al., 2024). Table 9 details the performance comparison between the standard baseline and M4D, quantifying the impact on both Accuracy and F1-score.

As shown in the table, M4D delivers a substantial improvement in operational performance under clean conditions, increasing the F1-score by +4.56% (from 79.72% to 84.28%). This gain is particularly critical in fraud detection, where the balance between precision and recall is paramount.

Under adversarial pressure, the superiority of M4D is evident in its F1-score stability. While the baseline suffers a catastrophic collapse under DeepFool (DF) attacks, dropping to an F1-score of 17.54%, M4D demonstrates significantly higher resilience with a +16.23% improvement (reaching 33.77%). Similarly, M4D outperforms the baseline under gradient-based attack (BIM) by +0.74% and maintains its high clean-level performance against C&W attacks (+4.56% improvement).

Table 8

Attack accuracy under different levels of attacker knowledge on the CICIoMT2024 dataset.

	Projc-only	Ent-only	Fully adaptive
Attack accuracy	0.99349	0.90375	0.53041

Table 9

Ablation study on credit card fraud dataset: Comparison of baseline vs. M4D across different attack scenarios. **Impact** denotes the percentage point difference (Δ) M4D achieves over the baseline.

Case	ACC			F1		
	DT	M4D	(Δ)	DT	M4D	(Δ)
Clean	78.99%	81.88%	+2.89%	79.72%	84.28%	+4.56%
BIM	71.74%	69.57%	-2.17%	72.34%	73.08%	+0.74%
DF	31.88%	26.09%	-5.79%	17.54%	33.77%	+16.23%
C&W	78.99%	81.88%	+2.89%	79.72%	84.28%	+4.56%
DT	99.28%	81.88%	-17.40%	99.35%	84.47%	-14.88%

A notable exception occurs under the DT attack, where the baseline exhibits an anomalously high performance spike (99.35%). However, excluding this outlier, M4D provides a more robust and consistent defense profile across the broader threat landscape. Crucially, these results were achieved using an off-the-shelf implementation of M4D originally designed for the MIoT domain, with the only modification being the adjustment of the input projection head to match the number of features in the fraud dataset. This absence of domain-specific hyperparameter tuning further validates that the proposed manifold entanglement offers intrinsic robustness that transfers effectively across disparate data modalities.

9. Conclusion

In this paper, we address domain shift and enhance generalizability against diverse unseen attacks in IIoT network environments by conceptualizing these challenges as robustness to evasion attempts. We propose a manifold-informed defense for DT-based NIDS, leveraging latent-space entanglement and cross-manifold mixup to maintain detection performance as traffic patterns evolve. Extensive experiments on the CIIoT2024 and ICU datasets show that our method balances clean accuracy with robustness to both white-box and black-box attacks, outperforming six benchmark defenses.

To further support and expand the generalization capability of the proposed method, future work will include a broader range of adversarial scenarios, including rigorous evaluation against adaptive attacks. Additional directions include exploring knowledge distillation to reduce training and testing time, as well as incorporating randomized or key-based entanglement mechanisms to thwart sophisticated white-box and adaptive adversaries capable of approximating manifold transformations.

CRedit authorship contribution statement

Mohammad Al-Fawa'reh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mohammed Kaosar:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration. **Jumana Abu-khalaf:** Writing – review & editing, Writing – original draft, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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